

E. G. S. PILLAY ENGINEERING COLLEGE

(Autonomous)

Approved by AICTE, New Delhi | Affiliated to Anna University, Chennai

Accredited by NAAC with A Grade | Accredited by NBA (CSE, ECE, IT)

NAGAPATTINAM – 611002



B.E. Biomedical Engineering

REGULATION - 2019

Full Time Curriculum and Syllabus

Total credit to be earned for the award of the Degree : 160 (Regular)
: 124 Lateral)

SEMESTER I									
Course Code	Course Name	L	T	P	C	Maximum Marks			Category
						CA	ES	Total	
Theory Course									
1901MA104	Engineering Mathematics –I (Linear Algebra, Calculus and Partial differentiation)	3	1	0	4	40	60	100	BSC
1901CH103	Chemistry for Biomedical Engineering	3	0	0	3	40	60	100	BSC
1901GEX03	Programming for Problem Solving	3	0	0	3	40	60	100	ESC
1901ENX01	English for Engineers	2	0	0	2	100	-	100	HSSC
Laboratory Course									
1901GEX52	Computer Programming Laboratory	0	0	2	1	50	50	100	BSC
1901GEX51	Engineering Intelligence I	0	0	2	1	50	50	100	HSSC
1901CHX51	Engineering Chemistry Laboratory	0	0	2	1	50	50	100	BSC
1901HSX51	Communication Skill Laboratory	0	0	2	1	100	0	100	HSSC
Total		11	1	8	16	470	330	800	
SEMESTER II									
Course Code	Course Name	L	T	P	C	Maximum Marks			Category
						CA	ES	Total	
Theory Course									
1901MA204	Engineering Mathematics–II (Calculus, Ordinary Differential Equations and Complex Variable)	3	2	0	4	40	60	100	BSC
1901PH202	Semiconductor Physics Optoelectronics	3	0	0	3	40	60	100	ESC
1901GEX01	Basic Electrical and Electronics Engineering	3	0	0	3	40	60	100	ESC
1901GEX02	Engineering Graphics	2	2	0	3	50	50	100	ESC
1901GE201	Engineering Exploration	2	0	0	2	40	60	100	ESC
Laboratory Course									
1901GE254	Computer Hardware and IT Essentials Laboratory	0	0	2	1	50	50	100	ESC
1901GE252	Engineering Intelligence - II	0	0	2	1	100	0	100	HSSC
1901GEX51	CAD Laboratory	0	0	2	1	50	50	100	ESC
1901GEX53	Basic Electrical and Electronics Engineering Laboratory	0	0	2	1	50	50	100	ESC
1901PHX51	Engineering Physics Laboratory	0	0	2	1	50	50	100	ESC
Total		13	4	10	20	510	490	1000	

SEMESTER III									
Course Code	Course Name	L	T	P	C	Maximum Marks			Category
						CIA	ES	Total	
Theory Course									
1901MA301	Engineering Mathematics III (Transforms and linear algebra engineering mathematics)	3	2	0	4	40	60	100	BSC
1902CS305	Object Oriented Programming and Data Structure	3	0	0	3	40	60	100	ESC
1902BM301	Fundamentals of Biochemistry	3	0	0	3	40	60	100	PC
1902BM302	Bio medical Circuits and Networks	3	2	0	4	40	60	100	PC
1902BM303	Biosensors and Measurements	3	0	0	3	40	60	100	PC
1902BM304	Human Anatomy and Physiology	3	0	0	3	40	60	100	PC
Laboratory Course									
1902BM351	Devices and Circuits Laboratory	0	0	2	1	50	50	100	PC
1902BM352	Biochemistry and Human Physiology Laboratory	0	0	2	1	50	50	100	PC
1902CS353	C++ and Data Structures Laboratory	0	0	2	1	50	50	100	ESC
1904GE351	Life Skills: Soft skill	0	0	2	1	100	-	100	EEC
Audit Course									
1901MCX02	Constitution of India	2	0	0	0	0	0	0	MC
Total		20	4	12	24	490	510	1000	
SEMESTER IV									
Course Code	Course Name	L	T	P	C	Maximum Marks			Category
						CIA	ES	Total	
Theory Course									
1901MA401	Probability Theory and Stochastic Processes	3	2	0	4	40	60	100	BSC
1902BM401	Biomechanics	3	2	0	3	40	60	100	PC
1902BM402	Basics of Pathology and Microbiology	3	0	0	3	40	60	100	PC
1902BM403	Control Systems	3	0	0	3	40	60	100	PC
1902BM404	Biomedical Instrumentation	3	0	0	3	40	60	100	PC
1902BM405	Digital Electronics and Integrated Circuits	3	0	0	3	40	60	100	PC
Laboratory Course									
1902BM451	Pathology and Microbiology Laboratory	0	0	2	1	50	50	100	PC
1902BM452	Biomedical Instrumentation Laboratory	0	0	2	1	50	50	100	PC
1902BM453	Analog and Digital Integrated Circuits Laboratory	0	0	2	1	50	50	100	PC
1904GE451	Life Skills: Verbal Ability	0	0	2	1	100	-	100	EEC
Audit Course									
1901MCX01	Environmental Science	2	0	0	0				MC
Total		20	04	08	23	490	510	1000	

SEMESTER V									
Course Code	Course Name	L	T	P	C	Maximum Marks			Category
						CIA	ES	Total	
Theory Course									
1902BM501	Biomedical Equipment's	3	0	0	3	40	60	100	PC
1902BM502	Medical Optics	3	0	0	3	40	60	100	PC
1902BM503	Microprocessor and its Applications	3	0	0	3	40	60	100	PC
1902BM504	Biomedical Digital Signal Processing	3	2	0	4	40	60	100	PC
1902BM505	Bio Process Control	3	0	0	3	40	60	100	PC
	Professional Elective - I	3	0	0	3	40	60	100	PE
Laboratory Course									
1902BM551	Biomedical Digital Signal Processing Laboratory	0	0	4	1	50	50	100	PC
1902BM552	Biosensors and Transducers Laboratory	0	0	4	1	50	50	100	PC
1904BM553	Microprocessor and Applications Laboratory	0	0	4	1	50	50	100	PC
1904GE551	Life Skills: Aptitude - I	0	0	2	1	100	-	100	EEC
Audit Course									
1901MCX03	Essence of Indian Traditional Knowledge	2	0	0	0	100	-	100	MC
Total		20	2	14	23	590	510	1100	
SEMESTER VI									
Theory Course									
1901BM601	Diagnostic and Therapeutic Equipment- I	3	0	0	3	40	60	100	PC
1902BM602	Analog and Digital Communication	3	0	0	3	40	60	100	PC
1902BM603	Biomaterials	3	0	0	3	40	60	100	PC
	Professional Elective – II	3	0	0	3	40	60	100	PE
	HSS Elective	3	0	0	3	40	60	100	HSS
Laboratory Course									
1902BM651	Diagnostic and Therapeutic Equipment Laboratory	0	0	4	1	50	50	100	PC
1902BM652	Analog and Digital Communication Laboratory	0	0	4	1	50	50	100	PC
1902BM653	Mini Project	0	0	4	1	50	50	100	EEC
1902BM654	Industrial Visit Presentation	0	0	2	1	100	-	100	EEC
1904GE651	Life Skills: Aptitude II GD	0	0	2	1	100	-	100	EEC
Total		15	0	16	20	550	450	1000	

SEMESTER VII									
Course Code	Course Name	L	T	P	C	Maximum Marks			Category
						CA	ES	Total	
Theory Course									
1902BM701	Diagnostic and Therapeutic Equipment- II	3	0	0	3	40	60	100	PC
1902BM702	Rehabilitation Engineering	3	0	0	3	40	60	100	PC
1901MGX07	Universal Human values and ethics	3	0	0	3	40	60	100	PC
	Professional Elective - III	3	0	0	3	40	60	100	PE
	Open Elective	3	0	0	3	40	60	100	OE
Laboratory Course									
1902BM751	Hospital Training	0	0	2	1	100	-	100	EEC
1904BM752	In-plant Training/ Internship Presentation	0	0	0	1	100	-	100	EEC
1904GE751	Life Skills : Comprehensive Viva	2	0	0	1	100	-	100	EEC
Total		20	0	2	18	590	410	1000	
SEMESTER VIII									
Theory Course									
1902BM801	Radiological Equipment's	3	0	0	3	40	60	100	PC
	Professional Elective – IV	3	0	0	3	40	60	100	PE
	Professional Elective – V	3	0	0	3	40	60	100	PE
Laboratory Course									
1904BM851	Project Work	-	-	14	07	50	50	100	EEC
Total		09	0	14	16	170	230	400	

Course Code	Course Name	L	T	P	C	Maximum Marks			Category
						CA	ES	Total	
PROFESSIONAL ELECTIVES – I									
1903BM001	Bio MEMS	3	0	0	3	40	60	100	PE
1903BM002	Nanotechnology in Medicine	3	0	0	3	40	60	100	PE
1903BM003	Bio Signal Processing and Systems	3	0	0	3	40	60	100	PE
1903BM004	Nuclear Medicine	3	0	0	3	40	60	100	PE
PROFESSIONAL ELECTIVES – II									
1903BM005	Regulatory Requirements in Pharmaceutical Industries	3	0	0	3	40	60	100	PE
1903BM006	Bio Analytical methods and Instruments	3	0	0	3	40	60	100	PE
1903BM007	Tele Health Technologies	3	0	0	3	40	60	100	PE
1903BM008	Medical Informatics	3	0	0	3	40	60	100	PE
PROFESSIONAL ELECTIVES – III									
1903BM013	Tissue Engineering and R-DNA technology	3	0	0	3	40	60	100	PE
1903BM014	Medical Imaging	3	0	0	3	40	60	100	PE
1903BM015	Pattern Recognition in Medicine	3	0	0	3	40	60	100	PE
1903BM016	3D Printing Technology	3	0	0	3	40	60	100	PE
PROFESSIONAL ELECTIVES – IV									
1903BM017	Rapid Prototyping	3	0	0	3	40	60	100	PE
1903BM018	Wearable Systems	3	0	0	3	40	60	100	PE
1903BM019	Disaster Managements	3	0	0	3	40	60	100	PE
1903BM020	Special Surgical Equipment	3	0	0	3	40	60	100	PE
PROFESSIONAL ELECTIVES – V									
1903BM021	Biometric Systems	3	0	0	3	40	60	100	PE
1903BM022	Robotics and Medicine	3	0	0	3	40	60	100	PE
1903BM023	Ergonomics	3	0	0	3	40	60	100	PE
1903BM024	Physiological modeling	3	0	0	3	40	60	100	PE
HSS ELECTIVES									
1901HS001	Innovation & Entrepreneurship Fundamentals	3	0	0	3	40	60	100	HSS
1901HS002	Intellectual Property Rights	3	0	0	3	40	60	100	HSS
1901HS003	Start Up Entrepreneurship	3	0	0	3	40	60	100	HSS
1901HS004	Business model Innovation	3	0	0	3	40	60	100	HSS
OPEN ELECTIVES OFFERED BY THE DEPARTMENT OF BIOMEDICAL ENGG									
1903BM009	Biomedical Engineering	3	0	0	3	40	60	100	OE
1903BM010	Design and development of biomedical systems	3	0	0	3	40	60	100	OE
1903BM011	Bio Process Modeling	3	0	0	3	40	60	100	OE
1903BM012	Instrumentation Engineering	3	0	0	3	40	60	100	OE